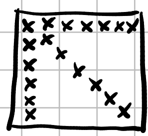
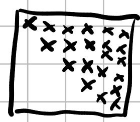
Reordering techniques for Cholesky / LU factorization of sparse matrices

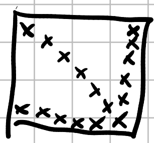
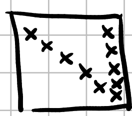
$Ax = b$ $\qquad$ A is SPD and SPARSE

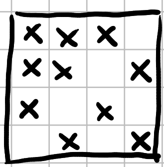
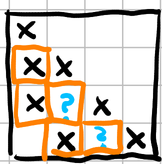
$A = R^T R$, $R =$ , $R^T =$ 

"Dream": A sparse $\overset{?}{\Longrightarrow}$ R is sparse

Counterexample: $A =$  $\quad R =$ 

Let's swap 1st and last row and column

$\rightsquigarrow B =$  $\rightsquigarrow R =$ 

$A =$  $\rightsquigarrow$ 

Def: $A \in \mathbb{R}^{n\times n}$, $J_i(A) = \min\{j : a_{ij} \neq 0\}$

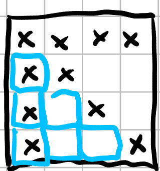
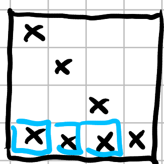
The "envelope" of A is

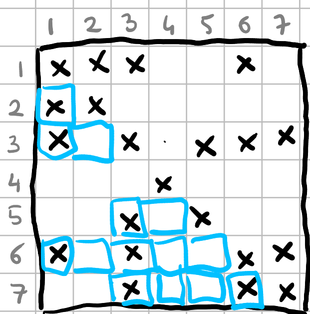
$$\mathrm{env}(A) := \{ (i,j) : J_i(A) \leq j < i \}$$

Theorem: $A \in \mathbb{R}^{n\times n}$ SPD, $A = LL^T$

Then $\ell_{ij} = 0$ whenever $(i,j) \notin \mathrm{env}(A)$ and $i > j$.

[The sparsity pattern of L is contained in the $\mathrm{env}(A)$].

Example 1:  

Example 2: 

$\rightarrow$ at most $14 + 7 = 21$ nnz in the Cholesky factor

To understand more precisely which entries of the Cholesky factor can be nonzero, we do an "elimination game" on the graph associated to the matrix

for $v = 1, 2, \dots, n$

connect all the neighbors of v

erase v and its corresp. edges

end

For the example above:

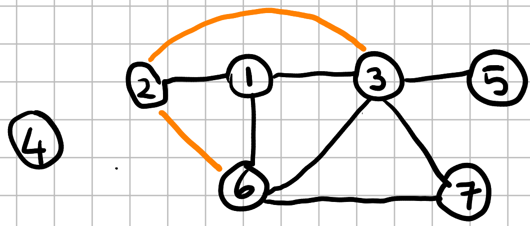
$$\begin{array}{c|ccccccc|}
 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
\hline
1 & \times & & & & & & \\
2 & \times & \times & & & & & \\
3 & \times & \times & \times & & & & \\
4 & & & & \times & & & \\
5 & & & \times & & \times & & \\
6 & \times & \times & \times & & \times & \times & \\
7 & & & \times & & \times & \times & \times \\
\hline
\end{array}$$

at most 18 nonzero entries

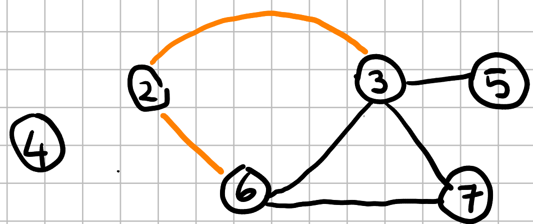
$\downarrow$

better than what the envelope of A predicts!

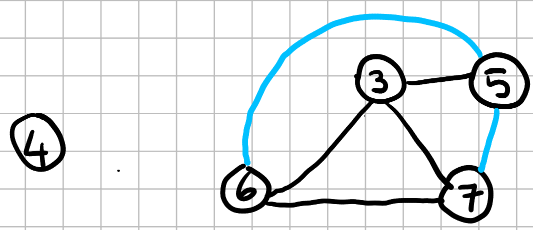
eliminating $v=1$



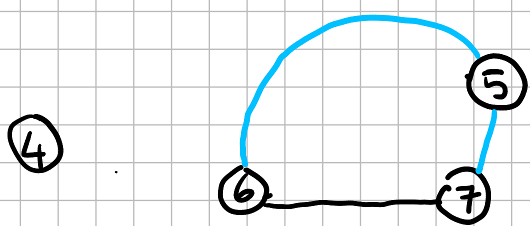
eliminating $v=2$



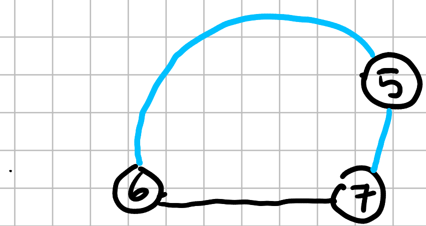
eliminate $v=3$



eliminate $v=4$



eliminate $v=5$

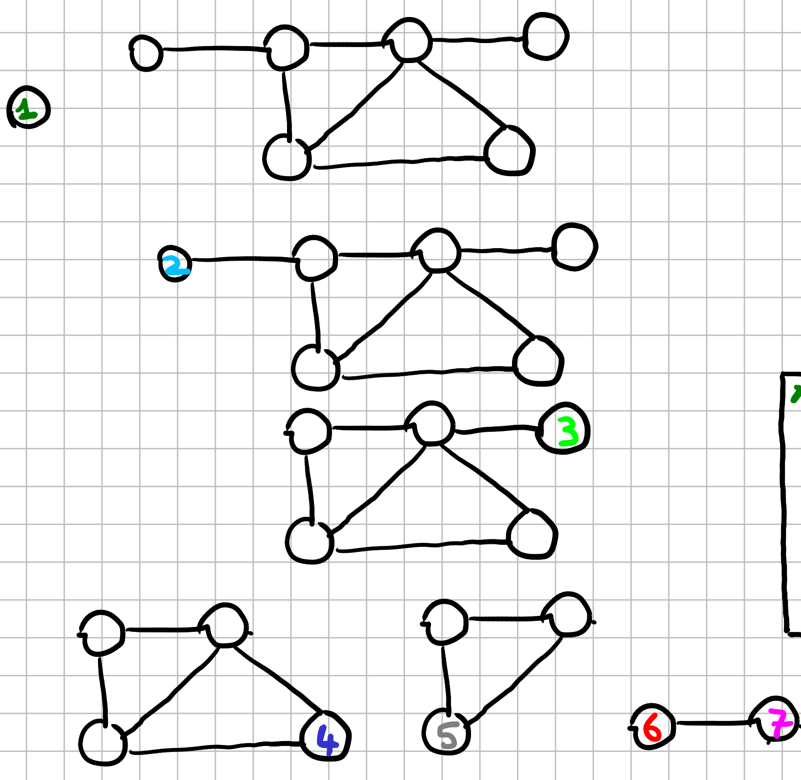


eliminate $v=6$

eliminate $v=7$

Ideas for reducing fill-in:

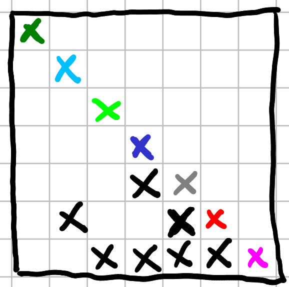
① (Approximate) minimum degree ordering



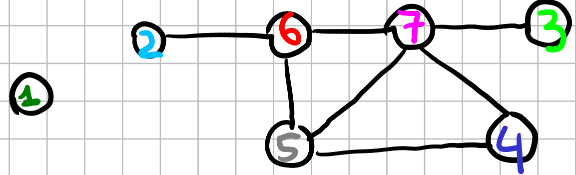
while (there are vertices)

take the one of min. degree and "eliminate" it in the sense of the elimination game seen before

end

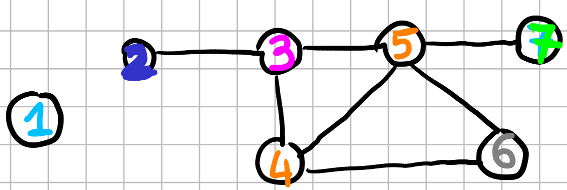


⇝ 14 nnz



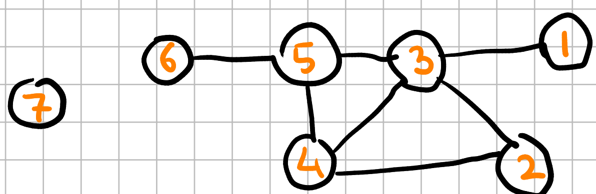
② Reverse Cuthill Mc-Kee ordering

start with one vertex (e.g. of min. degree), label the neighbors by ascending degree, then for each new vertex you just labeled, label its neighbors by ascending degree, and so on... then reverse the ordering.



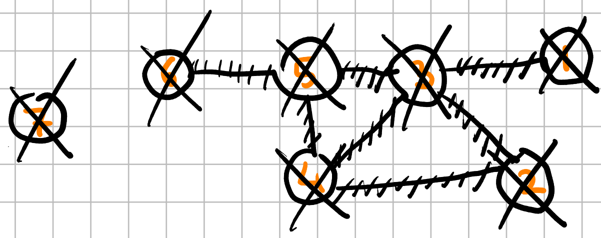
Cuthill - Mc-Kee ordering

(*)

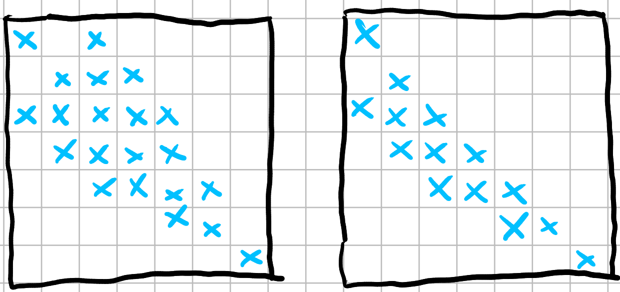


Reverse Cuthill Mc-Kee ordering

Reverse Cuthill Mc-Kee is faster than minimum degree ordering but if we want to know the sparsity pattern of the Cholesky factor, now we have to run the elimination game on (*)



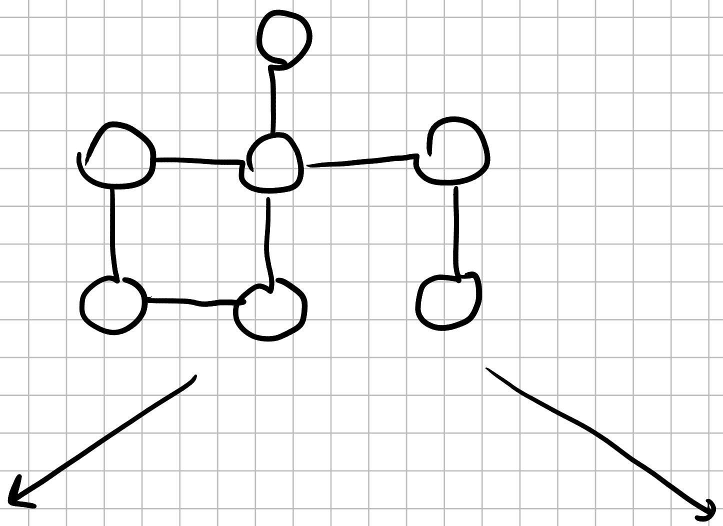
Here, no new edges are added $\Rightarrow$ sparsity pattern of the Cholesky factor is the same as the lower triangular part of the permuted A.



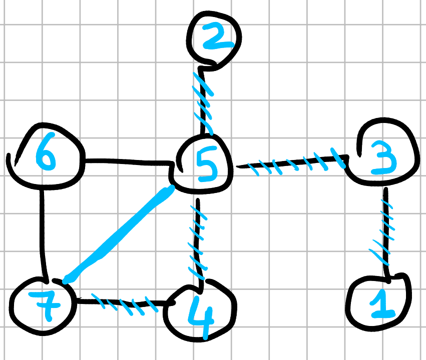
permuted matrix A

Cholesky factor

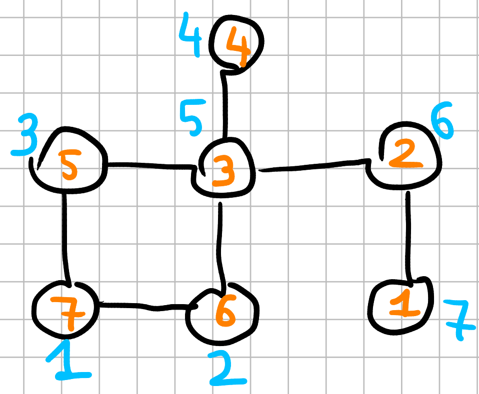
Example :

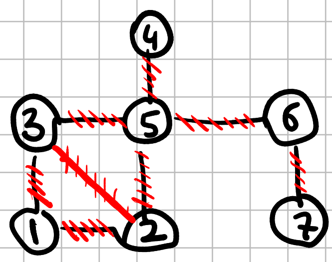


Minimum degree ordering



Reverse Cuthill McKee




4
3
5
6
1
2
7